

Hospital Dataset, Costing & AI Methodology

52K Hospital Dataset — Insurance Networks, PMJAY, Cost Estimation and Policy Analysis

Source	Reference / URL
Hospital Dataset	52K hospitals
ICICI Lombard Network	https://healthstatic.policybazaar.com/health-insurance/Network_Lists/ICICI_Lombard.Updated.pdf
Aditya Birla Network	https://healthstatic.policybazaar.com/health-insurance/Network_Lists/health-insuranceNetwork_ListsAditya_Birla.pdf
HDFC Network	https://joinditto.in/articles/content/files/2026/05/HDFC-Network-Hopital-List-ditto.pdf
Niva Bupa Network	https://healthstatic.policybazaar.com/health-insurance/Network_Lists/Niva_Bupa_(formerly_known_as_Max_Bupa).pdf
Star Health Network	https://healthstatic.policybazaar.com/health-insurance/Network_Lists/Star_Health1.pdf
PMJAY Hospital Dataset	https://github.com/AditiRoy171/pmjay_hospital_recommender/blob/main/synthetic_pmjay_dataset.csv
CGHS Cost / Rates	https://cghshospitals.com/rates
Room Cost Reference — Apollo Hospital Cost Breakdown	https://fittour.in/guides/apollo-hospital-cost-breakdown

Cost Estimation Engine — End-to-End Working

1. Procedure and city-tier benchmark foundation. Procedure costs are modeled across eight geographic tiers: **Metro 1, Metro 2, Large City 1, Large City 2, City 1, City 2, Town 1 and Town 2**. CGHS rates are used as a base benchmark, while procedure prices are researched/benchmarked across representative cities to create realistic low, mean and high ranges.

2. Procedure-specific estimated stay. The engine fixes an **estimated inpatient stay for each procedure** from the benchmark data. For example, Metro 1 Knee Replacement uses a **4-day estimated stay**. This stay is used when calculating accommodation charges.

3. Tier-wise room-cost estimation. Room costs are estimated from approximate **city-level prices** and mapped to the same eight geographic tiers. They are categorized into **General Ward, Twin Sharing and Single Private Room**. The FitTour — Apollo Hospital Cost Breakdown is used as a supporting reference for understanding approximate hospital room pricing. These are benchmark estimates, not exact hospital quotations.

4. Room charge calculation. The selected tier-wise daily room rate is multiplied by the procedure's estimated stay. For Metro 1 Knee Replacement: approximately **■2,500/day General, ■5,750/day Twin Sharing and ■12,000/day Private**. For 4 days this gives **■10,000, ■23,000 and ■48,000** respectively.

5. Hospital-specific procedure pricing. Exact tier + specialty + procedure data is selected first, with specialty/tier averages as fallbacks. Private hospitals use segment positions: Budget/Local 0.25, Standard Private 0.40, Established 0.50, Premium 0.65 and Luxury 0.80. A deterministic $\pm 5\%$ hash-based variation differentiates hospitals without randomness.

6. Final bill. Procedure charges + **20% doctor/specialist fees** + **10% medicines/diagnostics** + selected room rate $\times$ estimated stay = total estimated hospital bill.

7. Insurance impact. Policy exclusions, room-rent caps, proportionate deductions, procedure sublimits, deductibles, co-pay, sum-insured excess and modeled non-medical expenses are then applied to estimate insurer and patient shares.

Example: Knee Replacement in Metro 1

Benchmark: low **■50,000**, mean **■2,70,000**, high **■4,00,000**, estimated stay 4 days, with room rates of **■2,500/day general, ■5,750/day twin sharing and ■12,000/day private**.

Component	Calculation	Amount
-----------	-------------	--------

Procedure	Mean benchmark	■2,70,000
Doctor fees	20% × procedure	■54,000
Medicines & diagnostics	10% × procedure	■27,000
Twin-sharing room	■5,750 × 4 days	■23,000
Total	All components	■3,74,000

*The ■2,70,000 procedure amount is used only for demonstrating bill construction; the live engine uses hospital segment position and deterministic variation.

AI & Policy Analysis — Short Overview

AI is the reader, not the calculator. When a policy PDF is uploaded, the server extracts its text and sends it to **Google Gemini** with a strict JSON-schema prompt. Gemini identifies structured terms such as insurer, sum insured, room limit, ICU limit, co-pay, deductible, proportionate deduction, exclusions and procedure sublimits, while also returning source snippets and confidence levels.

Structured policy representation. The extracted information is normalized into a predictable JSON object. This object becomes the structured policy document consumed by the deterministic insurance-costing engine.

```
{
  "insurer": "string or null",
  "planName": "string or null",
  "policyType": "private | corporate | pmjay | esi | null",
  "sumInsured": "number (in rupees) or null",
  "roomLimit": {
    "type": "amount | percent | category | none",
    "value": "number or string or null"
  },
  "copay": "number (percentage) or null",
  "deductible": 0,
  "proportionateDeduction": null,
  "exclusions": ["string"],
  "sourceSnippets": {
    "fieldName": "exact source snippet"
  },
  "confidence": {
    "fieldName": "high | medium | low"
  }
}
```

What these fields represent: insurer and plan identify the policy; policyType distinguishes private/corporate/statutory schemes; sumInsured is normalized to rupees; roomLimit captures an amount, percentage or room category; copay and deductible represent cost-sharing rules; proportionateDeduction records whether room-cap breaches affect related medical charges; exclusions and sublimits capture restrictions; sourceSnippets preserve evidence from the uploaded document; and confidence records extraction confidence for each populated field.

Validation and safeguards. The JSON is validated with Zod. Invalid responses can be retried with validation feedback. Recognized statutory schemes such as PM-JAY and ESI can receive defined regulatory overrides. The confirmed policy is stored in MongoDB for use by the deterministic costing engine.

What the user sees. The Coverage Summary shows insurer/policy details, financial limits, cost-sharing terms, restrictions and sublimits, with document-stated versus scheme-standard provenance and clickable source snippets. Missing critical fields can be flagged for review.

Deterministic calculation flow. Policy PDF → Gemini extraction → structured JSON policy → dataset lookup → hospital pricing → bill breakdown → policy impact → insurer share + patient out-of-pocket. Financial arithmetic is performed by TypeScript/JavaScript formulas rather than by the LLM.

Gemini extraction controls. The prompt requires JSON-only output, prohibits guessing when a field is not explicitly stated, normalizes amounts to rupees and percentages to numbers, requires evidence snippets for populated fields, and assigns confidence. JSON response mode and zero temperature are used for structured, reproducible extraction.

Important distinction: Gemini interprets unstructured policy language; deterministic code performs the costing, room charges, proportionate deductions, co-pay, sublimits and final financial split.

Data Source Note

The supplied benchmark dataset contains tier-wise procedure ranges, mean costs, estimated stay days and three daily room-rate categories. The Metro 1 knee-replacement row contains █50,000 low, █4,00,000 high, █2,70,000 mean, 4 estimated stay days and █2,500 / █5,750 / █12,000 daily room rates for general / twin-sharing / private rooms.

Final Mock Database Look

The following screenshots show the final MongoDB mock database structure and sample records used in the project.

Database Link

<https://cloud.mongodb.com/v2/6a9d3db249e0752ff873ec0f#/explorer/6a9d3dfbfafddddd2d2997237>

Database Screenshot 1

```
_id: "pol_demo_hdfc"
confirmedByUser: false
insurer: "HDFC ERGO"
▶ insurerAliases: Array (1)
  planName: "Corporate Group Health Shield"
  policyType: "corporate"
  policyNumber: "HDFC/GRP/2026/98214"
  uin: "HDFHLGP21422V022021"
  policyStartDate: "2026-01-01"
  policyEndDate: "2026-12-31"
  zone: "Zone A"
  networkType: "all-network"
  tpa: "Vidal Health TPA"
▶ insuredPersons: Array (2)
  sumInsured: 5000000
▶ roomLimit: Object (2)
▶ icuLimit: Object (2)
  copay: 0
  nonNetworkCopay: 0
  deductible: 0
  proportionateDeduction: true
  restorationBenefit: true
  cumulativeBonus: 0
▶ exclusions: Array (2)
  hasOtherExclusions: false
▼ Show 13 more fields
```

Database Screenshot 2

```
_id: ObjectId('6abc1d0be1df00820528017a')
hospital_name: "St. Joseph Hospital"
nameNormalized: "st. joseph hospital"
hospital_type: "Private"
address: "Ram Camp Colony Opp Ivy Hospital Chandigarh Road, Hoshiarpur, Punjab, 146001"
city: "Hoshiarpur"
cityNormalized: "hoshiarpur"
▼ insurers: Array (1)
  0 : "Aditya Birla"
  insurersRaw: "Aditya Birla"
  rating: 0
▼ specialties: Array (7)
  0 : "Cardiology"
  1 : "Orthopedics"
  2 : "General Surgery"
  3 : "Neurology"
  4 : "Oncology"
  5 : "Gynecology"
  6 : "Urology"
tier: "City 2"
segment: "Standard Private"
__v: 0
createdAt: ISODate('2026-09-29T20:18:25.653+00:00') 29 September 2026 at 8:18:25 pm UTC
updatedAt: ISODate('2026-09-29T20:18:25.653+00:00') 29 September 2026 at 8:18:25 pm UTC
```

Database Screenshot 3

```
_id: ObjectId('6abc1d7a96b678bebl7bbc9a')
cost_type: "private"
tier: "Metro 1"
tierNormalized: "metro 1"
specialty: "Cardiology"
specialtyNormalized: "cardiology"
procedure: "Angiography"
procedureNormalized: "angiography"
low_cost: 5000
highest_cost: 210000
mean_cost: 25000
estimated_stay_days: 1
general_ward_cost_per_day: 2500
twin_sharing_cost_per_day: 5750
single_private_room_cost_per_day: 12000
__v: 0
createdAt: ISODate('2026-09-29T20:20:10.695+00:00') 29 September 2026 at 8:20:10 pm UTC
updatedAt: ISODate('2026-09-29T20:20:10.695+00:00') 29 September 2026 at 8:20:10 pm UTC
```